

Troubleshooting Guide: Three-Phase Inlet Separator — Emulsion-Driven Level Instability and Gas Valve Creep

Category: Separation Equipment / Process Chemistry & Control Interaction **Unit:** Three-Phase Inlet Separator (oil/water/gas, upstream gas processing facility) **Tools:** DCS trend/historian analysis with field lab (bottle test) verification — no steady-state process simulator required **Fluid Package:** Not applicable — no thermodynamic simulation was run (see Section 4, Step 1) **Symptom:** Liquid level swinging $\pm 10\text{--}20\%$ around normal, coupled with an unexplained gas outlet valve creeping open

Note on case type: Unlike Case Studies 1-5 (simulation-based diagnosis) and Case Study 6 (simulation-based predictive trending), this case was resolved entirely through **DCS historian analysis, instrumentation verification, and physical lab testing** — no process simulator or fluid package was involved. It's included to show that not every root-cause investigation requires a simulation model; sometimes the fastest and most defensible path is systematic data review + physical confirmation. This is also a good example of **cascading symptoms**: two seemingly separate control loops (liquid level, gas pressure) were actually both downstream effects of a single upstream cause.

1. Quick Reference

Item	Detail
Reported symptom	Liquid level swinging 40-70% (normally steady at 50%); gas outlet valve creeping open 3-7% (normally fully closed)
Initially unclear	Faulty instrumentation vs. poor control tuning vs. an actual process phenomenon inside the vessel
Actual root cause	A stable, slow-settling emulsion (rag layer) reduced effective separation volume; demulsifier dosing had recently been cut during a chemical optimization trial and was no longer sufficient for current crude properties
Fix	Restored demulsifier injection rate to its previous optimized value
Diagnostic tools	DCS historian trends; transmitter calibration checks; control valve stroke tests; field confirmation of rag layer; lab bottle test of emulsion stability
Prevention	New guideline requiring lab bottle tests before any future reduction in demulsifier dosage

2. Symptom

Two related but distinct symptoms appeared together:

- **Liquid level instability:** separator level, normally steady at **50%**, began **swinging between 40% and 70%**.
- **Gas outlet valve creep:** the gas outlet pressure control valve, which normally stayed **fully closed** (upstream pressure alone maintained separator pressure), started **creeping open 3-7%** as the controller hunted for a setpoint.
- **No alarms had fired yet** — but the correlated pattern across two different control loops signaled a genuine process upset rather than a nuisance/noise issue.

3. Why This Wasn't Immediately Obvious

Two separate control loops misbehaving at the same time could plausibly be explained by several very different fault classes, each requiring a different response:

1. **Faulty instrumentation** — a bad transmitter or valve, which would be a straightforward maintenance fix.
2. **Poor control tuning** — a controller retuning exercise, no physical vessel issue at all.
3. **An actual process phenomenon inside the vessel** — something changing the vessel's real separation behavior, which neither retuning nor instrument replacement would fix.

The task was explicit: figure out which of these it was, **and resolve it without taking the plant down** — ruling things out efficiently mattered as much as finding the answer.

4. Diagnostic Approach

Step 1 — Start with DCS historian trends

Historian trends were pulled for: - Separator level - Level control valve position - Gas outlet pressure and valve position - Inlet/outlet flows

No simulator needed for this step: this is a case where the fastest path to an answer is systematic trend review, not a process model. Simulation is a tool for testing a hypothesis quantitatively (as in Cases 1-5) or for predicting future behavior (Case 6) — it isn't always the first or best step. Here, the historian data itself was rich enough to start ruling out causes immediately.

Finding: Inlet flow was steady — this ruled out a **production-driven cause** (e.g., a slug of liquid or a rate change from upstream) as the trigger.

Step 2 — Verify instrumentation before suspecting process chemistry

Rather than jumping straight to “something's wrong inside the vessel,” the more easily-fixed and more common causes were checked first: - **Level transmitter calibration** - **Pressure transmitter health** - **Both control valves** — verified through **stroke tests**

Finding: All instrumentation checked out normal. This systematically eliminated fault classes 1 and 2 (faulty instrumentation, and by extension made poor tuning alone an unlikely explanation for two independent loops drifting together), and shifted the investigation **inside the vessel** — toward an actual process phenomenon.

Step 3 — Get field and lab confirmation

With instrumentation cleared, physical evidence was gathered directly: - **Field operators confirmed a persistent rag layer on the sight glass** — direct visual evidence of an

oil/water interface problem. - **Lab analysis (bottle test) confirmed a stable emulsion settling far slower than normal.**

Interpretation: This connected the two original symptoms into one mechanism: 1. The emulsion/rag layer **shrinks the effective separation volume** inside the vessel. 2. As the oil/water interface became unstable, the **level controller kept over-correcting the liquid outlet valve**. 3. Those liquid-side corrections **disturbed separator pressure**, which made the **gas outlet valve creep** in response.

What looked like two unrelated control loop problems was actually **one upstream cause propagating through two downstream control loops.**

Step 4 — Trace the emulsion back to a process chemistry change

Reviewing **chemical injection logs** revealed that **demulsifier dosing had recently been cut** as part of a chemical optimization trial. The new, lower dosing rate **wasn't sufficient to break the emulsion at current crude properties** — directly explaining why a stable emulsion had started forming when it hadn't before.

Quantitative Basis

- Separator level swung 40-70% against a normal steady 50% (alarm band 25-85%); gas outlet valve crept from fully closed (0%) to 3-7% open, hunting on a roughly 4-minute cycle.
- Bottle test at the reduced demulsifier rate (9 ppm): emulsion resolved to only 95% water-oil separation after 30 minutes.
- Bottle test at the previous rate (15 ppm): **>99% separation within 10 minutes** — confirming the dosing reduction, not the crude itself, was the driver.

5. Root Cause

A recent **reduction in demulsifier injection rate** (part of a chemical optimization trial) was insufficient to break the emulsion at the crude's current properties. The resulting **stable, slow-settling emulsion (rag layer)** reduced effective separation volume and destabilized the oil/water interface, which in turn caused the **level controller to over-correct**, which then **disturbed separator pressure and caused the gas outlet valve to creep open.**

6. Corrective Action

1. **Restored demulsifier injection rate to its previous optimized value.**
2. Allowed the emulsion to clear — resolved within a few hours.

7. Verification

- **Separator level held at 48-52% over the following 45 days**, versus the previous 40-70% swings.
- **Gas outlet valve remained fully closed (0-1%) throughout**, versus the previous 3-7% creep.
- **No further pressure fluctuations recorded.**
- All achieved **without a shutdown**, meeting the original task constraint.

8. Prevention / Long-Term Fix

- Established a **new chemical optimization guideline requiring lab bottle tests before any future reduction in demulsifier dosage**, so process chemistry changes are validated against actual crude behavior before being rolled out plant-wide.
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9. General Troubleshooting Checklist (Reusable)

Use this checklist when multiple, seemingly unrelated control loops on a three-phase separator start misbehaving together:

- ☐ Pull DCS historian trends for level, valve positions, pressures, and inlet/outlet flows — before assuming any single cause
- ☐ Check inlet flow stability first — rules production changes in or out quickly
- ☐ Verify transmitter calibration (level, pressure) and run control valve stroke tests before suspecting a process/chemistry cause
- ☐ If instrumentation checks out clean, look for a **shared upstream cause** connecting multiple misbehaving loops, rather than tuning each loop independently
- ☐ Get field confirmation (visual: sight glass, rag layer, interface appearance) — don't rely on instrumentation data alone for interface-related issues
- ☐ Run a **lab bottle test** to confirm emulsion stability/settling behavior quantitatively rather than relying on visual impression alone
- ☐ Review **chemical injection logs and recent dosing changes** — a chemistry change is easy to overlook if it happened as part of an unrelated “optimization” initiative
- ☐ Correct the chemistry (restore/adjust demulsifier rate) and confirm resolution against the ORIGINAL symptoms (level stability, valve position, pressure), not just visual clearing of the rag layer
- ☐ Add a validation step (e.g., bottle testing) to the change-management process for any future dosing reduction, so optimization trials don't silently reintroduce the same failure mode

10. Key Takeaway

When two different control loops start misbehaving at the same time, don't tune them separately — look for a **single upstream cause** creating both symptoms. Rule out instrumentation first (it's the fastest, most common, and most fixable explanation), but if everything checks out clean, be willing to go physical: a sight glass observation and a lab bottle test can find what historian trends and stroke tests can't. And when the root cause turns out to be a recent “optimization” change, remember that any dosing/setpoint change to process chemistry should be validated against current feed properties before rollout — not assumed safe because it worked historically.

Related Concepts / Tags

three-phase-separator emulsion rag-layer demulsifier level-instability valve-creep DCS-historian bottle-test control-loop-interaction chemical-optimization process-chemistry instrumentation-verification

This guide is derived from a real field troubleshooting case. Values and thresholds shown are specific to the reported case and should be validated against your own unit's design basis before applying.